

Misr University for Science and Technology

Intelligent Plant Disease Detection Using Deep Learning

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Abstract

Plant diseases pose a serious threat to agricultural productivity and food security worldwide. Early detection of plant diseases is essential to reduce crop losses and improve yield quality. Traditional disease detection methods rely on manual inspection, which is often time-consuming, costly, and subject to human error.

This project presents an AI-based plant disease detection system that utilizes deep learning techniques to automatically identify and classify plant diseases from plant leaf images. The proposed system employs convolutional neural networks (CNNs) and image processing methods to achieve accurate and efficient disease diagnosis. Multiple pre-trained deep learning architectures were evaluated to select the most suitable model based on performance and efficiency.

The current phase of the project focuses on the design and development of the AI-based disease detection module. The system is designed to support future integration with environmental monitoring components, including temperature, humidity, soil moisture, and light intensity sensors, to enhance disease analysis and decision-making.

The experimental results demonstrate the effectiveness of deep learning models in plant disease classification, highlighting the potential of the proposed system to support smart agriculture applications and early disease detection.

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CHAPTER 1

Introduction

This chapter introduces the proposed plant disease detection system and provides an overview of the project concept, motivation, and objectives. It highlights the importance of early plant disease diagnosis and the role of artificial intelligence in modern agricultural applications. The chapter also presents the project scope and development phases, establishing a clear foundation for the system design and methodology discussed in subsequent chapters.

1 INTRODUCTION

1.1 Brief Overview of the Project

Agriculture plays a vital role in ensuring food security and economic stability worldwide. One of the major challenges facing the agricultural sector is the spread of plant diseases, which can significantly reduce crop yield and quality if not detected at early stages. Traditional methods of plant disease detection rely heavily on manual inspection by experts, which is time-consuming, costly, and prone to human error.

This project proposes an intelligent plant disease detection system based on artificial intelligence (AI) techniques. The system focuses on analyzing plant leaf images using deep learning models to automatically identify and classify plant diseases. By leveraging convolutional neural networks (CNNs) and image processing techniques, the proposed system aims to provide fast and accurate disease diagnosis.

In addition to image-based analysis, the system is designed to incorporate environmental data such as temperature, humidity, soil moisture, and light intensity in later stages of the project. These factors play a significant role in plant health and disease development. However, the current phase of the project primarily focuses on the AI-based disease detection module, while the hardware implementation is planned for a future phase.

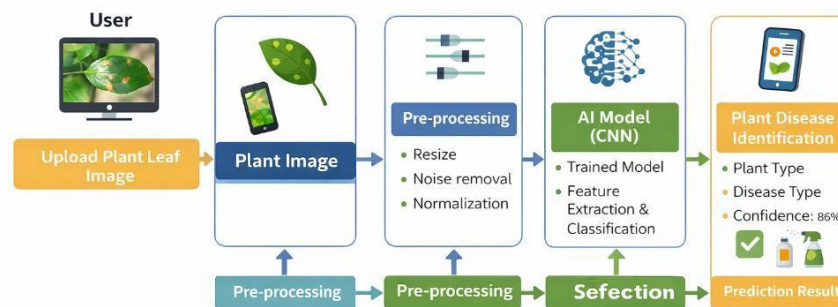


Figure 1.1: General overview of the proposed AI-based plant disease detection system.

1.2 Significance

Early detection of plant diseases is essential to prevent their spread and minimize agricultural losses. An automated disease detection system can support farmers and agricultural specialists by providing timely and reliable diagnosis, reducing dependency on manual inspection.

The significance of this project lies in its integration of artificial intelligence with smart agriculture concepts. By utilizing deep learning models for image-based disease detection, the system offers a scalable and cost-effective solution that can be adapted to various crop types. Moreover, the planned integration of environmental monitoring in future stages enhances the system's potential to support informed decision-making and sustainable agricultural practices.

1.3 Background

Plant diseases are caused by various factors, including fungi, bacteria, viruses, and unfavorable environmental conditions. Changes in temperature, humidity, soil moisture, and light intensity can create suitable environments for disease development and progression. Understanding these factors is crucial for effective disease management.

Recent advancements in artificial intelligence, particularly in deep learning and computer vision, have enabled significant improvements in image-based classification tasks. Convolutional neural networks (CNNs) have demonstrated high performance in recognizing visual patterns and have been widely applied in agricultural applications such as plant disease detection. These techniques provide a foundation for developing intelligent systems capable of analyzing plant images with high accuracy.

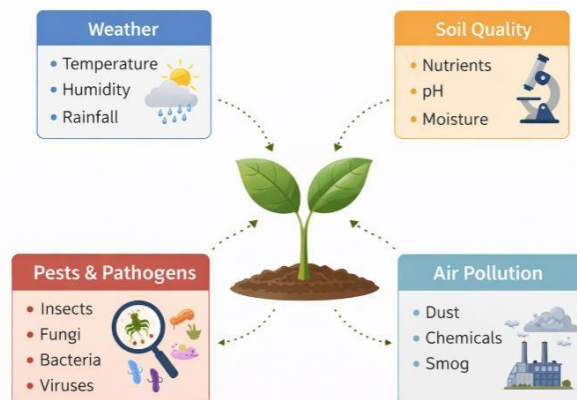


Figure 1.2: Environmental factors influencing plant health and disease development.

1.4 Objectives

The main objectives of this project are as follows:

- To design an AI-based system capable of detecting and classifying plant diseases from leaf images.
- To evaluate and compare different deep learning architectures for disease detection.
- To improve disease diagnosis accuracy through image preprocessing and model optimization techniques.
- To propose a system framework that supports future integration with environmental monitoring sensors.

1.5 Scope

The scope of this project is limited to the development and evaluation of an AI-based disease detection system using image data. The current phase focuses on model training, testing, and performance analysis. Hardware implementation, including the use of sensors and embedded systems, is considered part of future work and will be addressed in subsequent project phases.

The system is intended for research and educational purposes and is not designed to replace professional agricultural expertise. It serves as a decision-support tool to assist in early disease detection.

1.6 Project Phases

The project is divided into three main phases:

- ✚ **Phase 1: System Design and AI Model Development**
 - Data collection and preprocessing
 - Deep learning model selection and training
 - Model evaluation and performance analysis
- ✚ **Phase 2: Interface Design and Application Development**
 - Conceptual UI/UX design for the mobile application
 - Conceptual UI/UX design for the web-based dashboard
 - Visualization of plant disease detection results
- ✚ **Phase 3: Hardware Implementation and System Integration**
 - Integration of camera and environmental sensors
 - Deployment on an embedded platform
 - Real-time monitoring and system validation



Figure 1.3: Project phases for system development and implementation.

CHAPTER 2

BACKGROUND AND RELATED WORK

This chapter presents the background and related work associated with plant disease detection using artificial intelligence and intelligent systems. It reviews recent research studies that employ deep learning models, datasets, software tools, and different methodological approaches for plant disease classification, detection, and segmentation. The chapter analyzes each study individually to highlight similarities, differences, and limitations, providing a clear understanding of the current state of the field and identifying research gaps that motivate the proposed system.

2 BACKGROUND AND RELATED WORK

2.1 Background

Plant diseases significantly affect agricultural productivity and crop quality, leading to economic losses and food security challenges. These diseases are caused by pathogens such as fungi, bacteria, and viruses, in addition to adverse environmental conditions. Early detection of plant diseases is critical to prevent their spread and to enable timely intervention.

Environmental parameters including temperature, humidity, soil moisture, and light intensity influence plant health and disease progression. Variations in these factors can create favorable conditions for disease development. Understanding their impact is essential for designing intelligent plant disease detection systems.

Recent advances in artificial intelligence, particularly deep learning and computer vision, have enabled automated plant disease detection systems. Convolutional Neural Networks (CNNs) and encoder-decoder architectures have shown strong performance in analyzing plant leaf images, making them suitable for real-world agricultural applications.

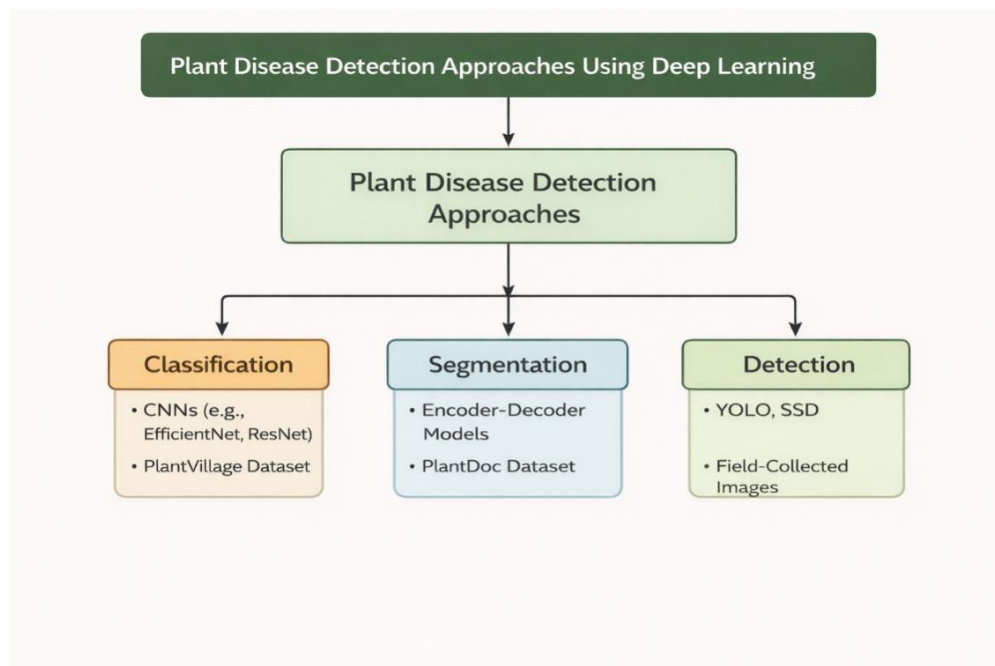


Figure 2.1: Deep Learning Approaches for Plant Disease Detection

2.2 Related Work

This section reviews recent studies related to plant disease detection using deep learning and intelligent systems. Each study is analyzed individually to highlight differences in models, datasets, methodologies, and system objectives.

Paper 1: An Analysis of the Accuracy of Photo-Based Plant Identification Applications on Fifty-Five Tree Species

Model and Dataset:

This study evaluated six widely downloaded photo-based plant identification applications (PictureThis, iNaturalist, Plant Identification, PlantNet, LeafSnap, and PlantSnap). The evaluation was conducted using 440 field photographs (4 bark and 4 leaf images for each species) representing 55 common tree species in New Jersey forests and urban landscapes. Images were captured using smartphone cameras and tested under consistent conditions without GPS metadata.

System and Functionality:

The applications perform plant identification by analyzing uploaded leaf or bark images using machine learning and image recognition techniques. Most apps provide a primary identification along with additional suggestions. Some platforms (e.g., iNaturalist, PlantNet, PlantSnap) include community-based verification features, although only automated AI-based suggestions were evaluated in this study.

Software:

The apps rely on image recognition algorithms trained on large-scale botanical image datasets. Identification is probability-based, often using neural network models trained in controlled environments. However, the study evaluated performance outside training environments using real, non-curated field images.

Methodology:

Each of the 440 images was uploaded individually to all six applications. Identification accuracy was measured at both genus and species levels for primary identification and suggested results. Accuracy percentages were calculated separately for bark and leaf images. A chi-squared (χ^2) test was conducted to verify repeatability of app outputs across multiple runs.

Key Features and Benefits:

Results showed that leaf images were significantly more accurately identified than bark images across all apps. PictureThis achieved the highest overall accuracy (97.27% genus-level accuracy for leaves and 83.86% species-level accuracy). iNaturalist showed strong genus-level performance and high suggestion accuracy. Overall, apps were more reliable at genus-level identification than species-level identification. The study concluded that while these applications cannot replace traditional field identification methods, they are useful support tools for education, early training, and community science projects.[1]

Paper 2: Evaluating Plant Disease Detection Mobile Applications: Quality and Limitations

Model and Dataset:

This study did not propose a new deep learning model; instead, it evaluated existing mobile applications for plant disease detection. A total of 606 applications were identified from three major app stores (Google Play, Apple App Store, and Microsoft Store), and 17 relevant apps were selected after applying PRISMA-based screening and inclusion/exclusion criteria. Many of these applications rely on AI models trained on image datasets such as PlantVillage, which contains over 54,000 labeled images across multiple crops and disease classes.

System and Functionality:

The evaluated applications perform plant disease detection by allowing users to upload or capture images of plant leaves. The systems analyze images using AI/ML or deep learning techniques for plant identification and disease diagnosis. The most comprehensive app identified in the study was Plantix—your crop doctor, which supports automatic plant identification, disease detection, treatment recommendations, and community support. However, most apps lacked advanced functionalities such as automatic infected area visualization and disease severity estimation.

Software:

The study developed and applied a structured App Rating Scale adapted from the Mobile App Rating Scale (MARS). The evaluation covered aesthetics, usability, performance and efficiency, functionality, transparency, subjective quality, and perceived user impact. Three independent raters assessed the apps, and inter-rater and intra-rater reliability were validated using ICC and Cronbach's alpha, achieving high internal consistency (α (0.97 =

Methodology:

A systematic search strategy following PRISMA guidelines was used. After removing duplicates ($n = 12$) and applying screening criteria, 17 apps were selected for detailed evaluation. Each app was independently rated by three expert raters using a 1–5 Likert scale across multiple subscales. Statistical validation included inter-rater reliability (ICC = 0.869) and internal consistency testing. User reviews from app stores were also analyzed to understand strengths and limitations.

Key Features and Benefits:

Most apps demonstrated good usability and performance but performed poorly in AI-based advanced functionalities. Approximately 88% of apps supported automatic plant identification and disease detection. However, only 23.52% supported disease severity estimation, 5.88% visualized infected areas, and 11.76% provided expert/community support. Among all apps, Plantix–your crop doctor achieved the highest overall rating (mean score = 4.56) and was identified as the most comprehensive solution. Nevertheless, the study concluded that most existing apps depend heavily on controlled datasets and lack integration with environmental data or IoT systems, limiting real-world applicability.[2]

Paper 3: Non-Linear Analysis in Post-Buckling Regime of a Tilt Rotor Composite Wing Structure Using Detailed Model and Robust Loading Approach

Model and Dataset:

This study does not use a machine learning dataset; instead, it focuses on a high-fidelity structural finite element (FE) model of a composite tilt-rotor wing. A detailed finite element model (DFEM) was developed in MSC Nastran, including over 1.2 million shell elements and approximately 1.27 million nodes. The wing structure is primarily composed of composite materials (fabric CFRP, unidirectional CFRP, and sandwich honeycomb structures), with some metallic components made of aluminum alloy 7050-T7451. The model represents the complete wing integrated within the full aircraft configuration.

System and Functionality:

The system performs non-linear structural analysis in the post-buckling regime to evaluate the wing's behavior under design load conditions. Unlike traditional approaches that avoid buckling up to ultimate load, this methodology allows controlled buckling while ensuring no structural failure occurs before ultimate load. The model includes inertia relief to simulate real flight conditions (free-free structure), aerodynamic loads, propulsion forces, fuel mass distribution, and balanced aircraft loading.

Software:

The structural analysis was performed using MSC Nastran (SOL 400) with large displacement non-linearity activated (PARAM, LGDISP, 1). Custom Python scripts and HDF5 data extraction were developed to generate equivalent balanced loads and replace inertia relief for non-linear analysis. Additional modeling tools included PCOMP/MAT8 for composite materials, PSHELL/MAT1 for isotropic parts, and RBE3/MPCFORCE procedures for load distribution.

Methodology:

A novel numerical procedure was developed to extend inertia relief functionality to non-linear post-buckling analysis. First, a linear static analysis with inertia relief was performed to extract balanced loads. These loads were converted into equivalent FORCE/MOMENT entries using external Python scripts. Large RBE3 elements were replaced with equivalent distributed loads to improve computational efficiency. Contact conditions were introduced to prevent unrealistic instability modes. The methodology enabled step-by-step non-linear implicit analysis under realistic flight loading conditions.

Key Features and Benefits:

The proposed approach enables accurate simulation of post-buckling behavior while preserving structural integrity. It allows exploitation of composite material strength beyond initial buckling, leading to potential weight savings compared to conventional conservative design approaches. The results demonstrated that differences between linear and non-linear regimes are significant and that post-buckling capability is critical for achieving optimized lightweight design in next-generation tilt-rotor aircraft.[3]

Paper 4: AI-IoT Based Smart Agriculture Pivot for Plant Disease Detection and Treatment

Model and Dataset:

ResNet50 was employed for plant disease classification using the New Plant Diseases Dataset (NPDD) containing 25,940 images across 11 classes.

System and Functionality:

The system integrates AI with IoT components, including sensors, cameras, Raspberry Pi, and actuators. Upon disease detection, the system automatically applies fertilizers or pesticides and provides guidance through a mobile application.

Software:

Image augmentation techniques were applied, and the trained model was integrated into a smart pivot system and a mobile application for real-time disease diagnosis.

Methodology:

Captured leaf images were augmented and processed before classification. ResNet50 performance was compared against other CNN architectures.

Key Features and Benefits:

The system achieved high classification accuracy and demonstrated successful real-time detection and automated treatment, offering a low-cost alternative to robotic solutions.[4]

Paper 5: AI and IoT-Powered Edge Device Optimized for Crop Pest and Disease Detection

Model and Dataset:

A lightweight CNN model, Tiny-LiteNet, was trained on PlantVillage images and field-acquired pest and disease images.

System and Functionality:

The proposed system is a portable AI-powered IoT device designed for real-time pest and disease detection in resource-limited environments.

Software:

The system integrates edge computing software optimized for Raspberry Pi 5, with support for GSM/GPRS communication.

Methodology:

Data augmentation and cross-validation were used to evaluate performance. Benchmarking was conducted against larger CNN architectures.

Key Features and Benefits:

Tiny-LiteNet achieved 98.6% accuracy with a compact model size of 1.2 MB and low inference latency, making it suitable for portable agricultural applications.[5]

Paper 6: Paddy Disease Detection and Classification Using Computer Vision Techniques

Model and Dataset:

The study used CNN models (ResNet, VGG, MobileNet), Vision Transformer (ViT), and YOLOv8. The Paddy Doctor dataset containing 16,225 images with annotated bounding boxes was employed.

System and Functionality:

The system supports both disease classification and detection, enabling identification of single and multiple diseases in paddy crops.

Software:

The models were deployed within a mobile application supported by cloud-based services for real-time diagnosis.

Methodology:

Classification and detection models were evaluated using accuracy, F1-score, and mAP metrics.

Key Features and Benefits:

ViT achieved the highest classification accuracy, while YOLOv8 enabled multi-disease detection, demonstrating flexibility in disease diagnosis approaches.[6]

Paper 7: Role of Internet of Things and Deep Learning Techniques in Plant Disease Detection and Classification – A Focused Review

Model and Dataset:

This review analyzed multiple deep learning architectures, including CNNs and lightweight models, across datasets such as PlantVillage and PlantDoc.

System and Functionality:

The study emphasized IoT-enabled systems that combine real-time sensing with deep learning-based disease detection.

Software:

Various deep learning and IoT platforms were reviewed, highlighting the role of mobile and embedded systems.

Methodology:

Over 150 research papers were systematically reviewed to compare standalone deep learning systems with IoT-integrated approaches.

Key Features and Benefits:

The integration of IoT and deep learning improves detection accuracy and sustainability but introduces challenges related to power consumption and connectivity.[7]

Paper 8: A Mobile-Based System for Detecting Plant Leaf Diseases Using Deep Learning

Model and Dataset:

A CNN-based model was trained on a large custom dataset containing 96,206 images from multiple public sources.

System and Functionality:

The system enables real-time plant disease detection through a smartphone application.

Software:

The CNN model was implemented using Keras and TensorFlow and deployed on Android using OpenVINO.

Methodology:

Image preprocessing and augmentation were applied to improve robustness. Performance was evaluated based on accuracy and inference time.

Key Features and Benefits:

The system achieved approximately 94% accuracy and provided fast predictions suitable for mobile deployment.[8]

Paper 9: APEIOU Integration for Enhanced YOLOv7: Efficient Plant Disease Detection

Model and Dataset:

An enhanced YOLOv7 architecture was trained on combined PlantVillage and PlantDoc datasets.

System and Functionality:

The system focuses on efficient detection of small and complex plant disease targets.

Software:

Custom training strategies and attention mechanisms were implemented to improve detection performance.

Methodology:

Structural improvements and a novel APEIOU loss function were introduced to enhance detection accuracy.

Key Features and Benefits:

The enhanced YOLOv7 model achieved high mAP and precision while maintaining a lightweight structure suitable for agricultural applications.[9]

2.3 Summary of Related Work

The reviewed studies collectively demonstrate the effectiveness of deep learning techniques for plant disease detection across various system designs and deployment scenarios. Classification-based CNN models provide efficient and accurate disease identification, while encoder–decoder segmentation approaches enhance precision by localizing diseased regions before classification. YOLO-based detection models enable real-time disease localization, particularly suitable for edge and embedded systems.

Several studies emphasize the importance of dataset diversity, highlighting that models trained on combined real-world and curated datasets achieve better generalization under varying environmental conditions. Edge-optimized and lightweight architectures such as MobileNet and customized YOLO variants support real-time inference with reduced computational and energy requirements, making them suitable for smart agriculture applications.

Despite the promising results, common limitations remain across the literature, including dataset imbalance, sensitivity to real-world noise, computational complexity of advanced architectures, and limited integration of environmental sensing data. These challenges motivate the proposed system, which aims to balance accuracy, efficiency, and scalability while supporting future integration with hardware-based monitoring and user-friendly mobile and web interfaces.

CHAPTER 3

Methodology

This chapter describes the proposed methodology used for plant disease detection and classification. It outlines the overall system workflow, dataset preparation, deep learning model training process, and software development strategy. Hardware implementation and security considerations are intentionally excluded at this stage, as they are planned for the second project phase.

Methodology

3.1 Methodology

This section presents the overall workflow of the proposed system, starting from image acquisition to disease classification. The system focuses on an AI-based image classification approach trained on publicly available plant disease datasets. Images are preprocessed, augmented, and fed into a trained deep learning model, which outputs the predicted disease class.

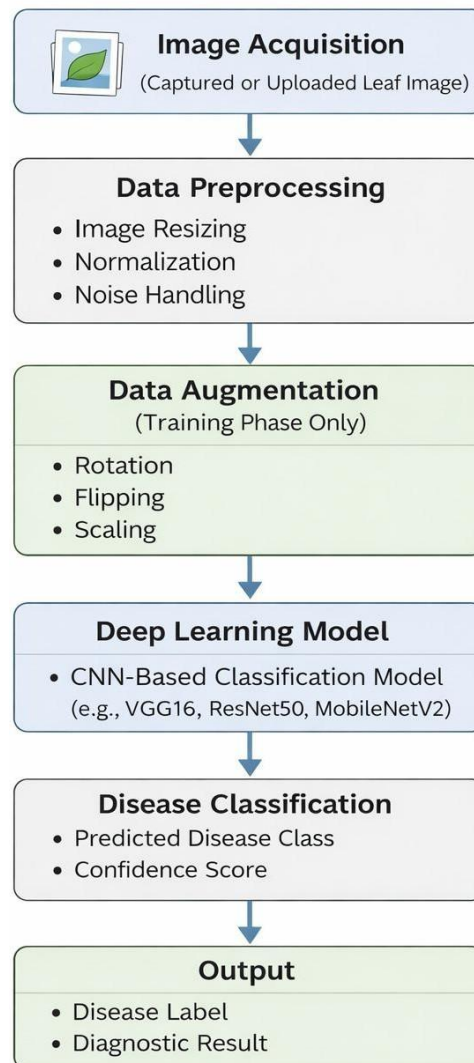


Figure 3.1: Overall system workflow for plant disease detection and classification.

3.2 Component Selection

This section explains the datasets and software components selected for the first phase of the project.

3.2.1 Datasets Description

- **PlantVillage Dataset:** Used as the primary curated dataset containing labeled images of healthy and diseased plant leaves under controlled conditions.
- **RiceLeafs Dataset:** Used to represent crop-specific disease cases and improve model diversity.

The two datasets were merged to increase data diversity and enhance model generalization.

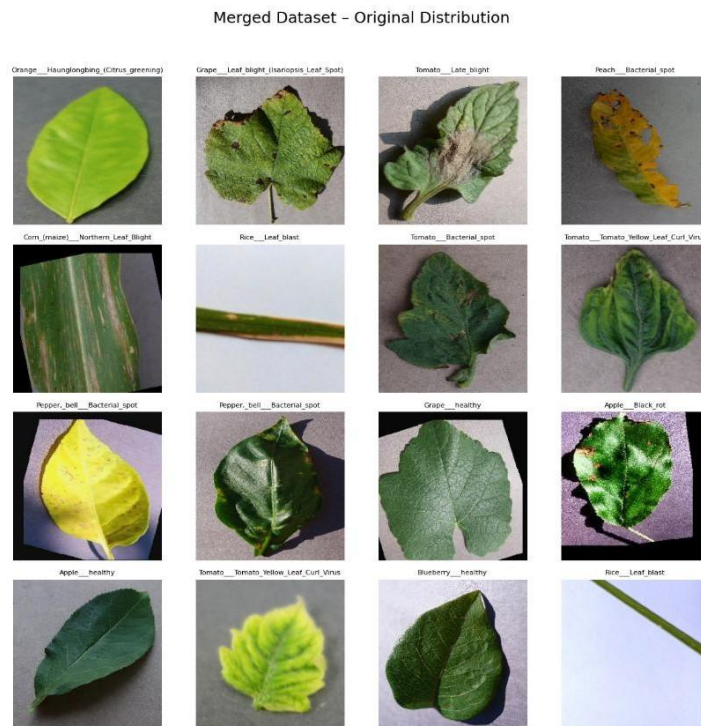


Figure 3.2: Sample images from PlantVillage and RiceLeafs datasets.

3.2.2 Data Preprocessing and Augmentation

During preprocessing, background removal was applied to minimize the effect of complex backgrounds and improve feature extraction by focusing on the plant leaf region. This step enhances model robustness under varying real-world imaging conditions. Data augmentation was applied only during the training phase to improve generalization, while inference follows the same preprocessing pipeline without augmentation

3.3 Comprehensive

This section describes the deep learning models selected for plant disease classification, the unified training configuration, and a high-level comparison of model performance. Three convolutional neural network architectures were evaluated to balance accuracy, computational complexity, and deployment feasibility.

3.3.1 Selected CNN Architectures

Three well-established CNN architectures were selected for experimentation: - **VGG16-BN**: A deep convolutional network known for its stable training behavior and strong feature extraction capability. - **ResNet50**: A residual network designed to address the vanishing gradient problem in deep architectures. - **MobileNetV2**: A lightweight architecture optimized for efficiency and mobile or edge deployment.

These models were chosen to represent different design philosophies, ranging from high-capacity networks to resource-efficient architectures.

3.3.2 Training Configuration (Common Setup)

To ensure a fair comparison, all models were trained using the same experimental setup. The merged PlantVillage and RiceLeafs datasets were used with identical preprocessing and augmentation techniques. The dataset was split into training, validation, and testing subsets. All models employed the same loss function and optimizer to maintain consistency across experiments.

3.3.3 Model Performance Overview

A summarized qualitative assessment of each model is presented below:

- **VGG16-BN**: Demonstrated strong convergence and stable learning behavior, achieving high classification accuracy across both datasets.
- **ResNet50**: Achieved competitive performance through deep residual learning, with improved feature representation at the cost of increased computational complexity.
- **MobileNetV2**: Offered efficient training and inference with lower computational requirements, making it suitable for future mobile and embedded deployment.

The detailed quantitative results and in-depth evaluation metrics are deferred to the results and discussion phase.

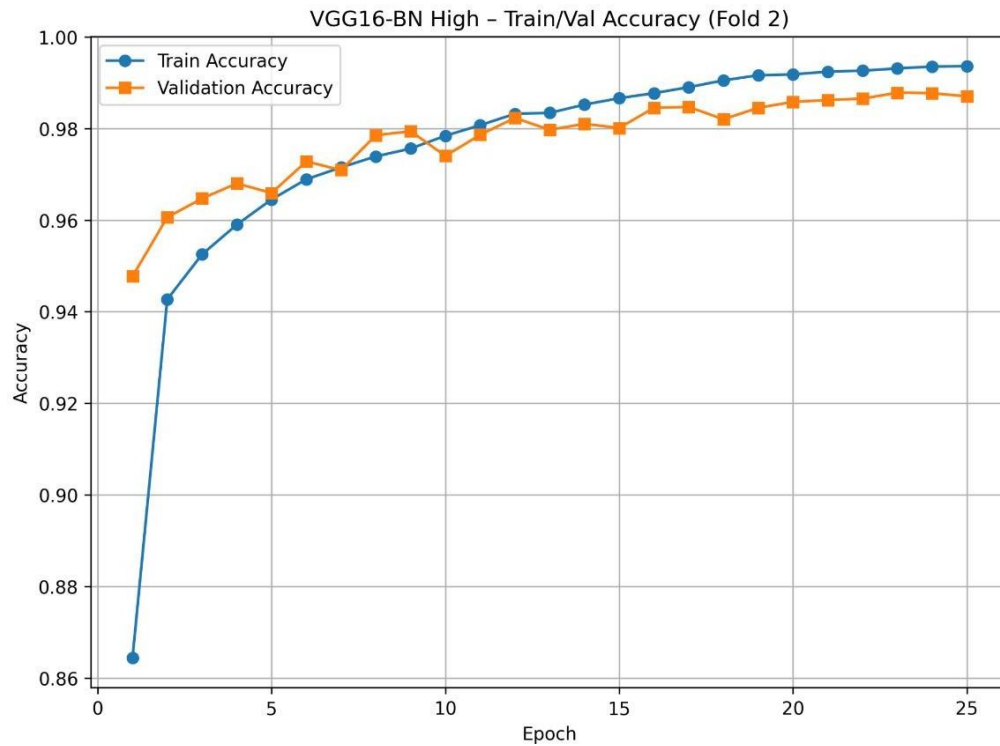


Figure 3.3: Training and validation accuracy curves of the VGG16-BN model.

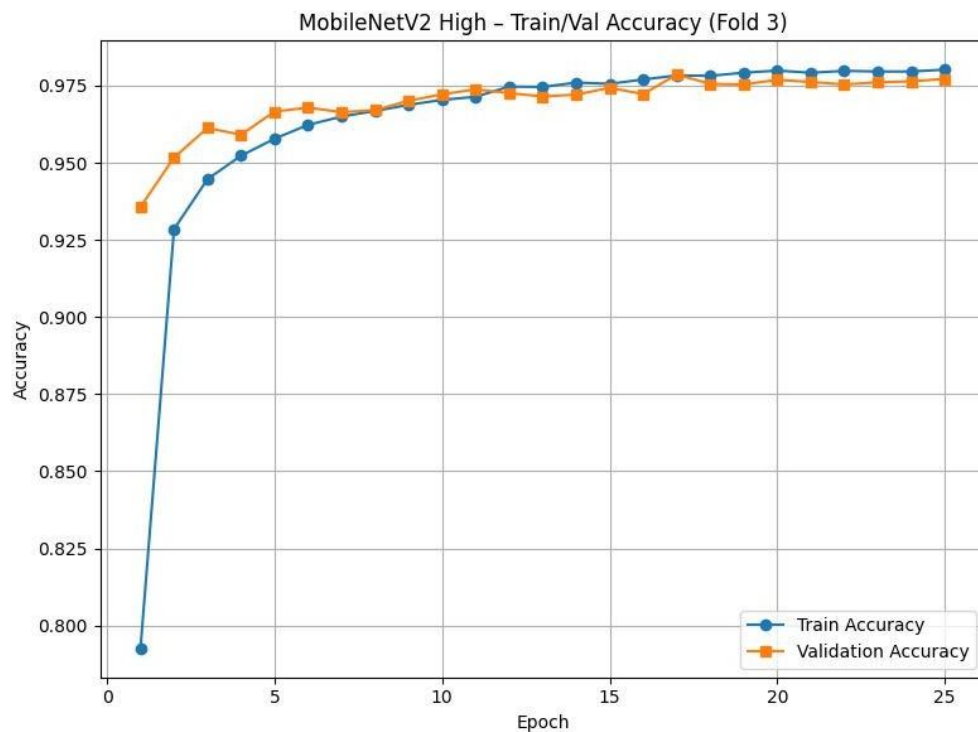


Figure 3.4: Training and validation accuracy curves of the MobileNetV2 model.

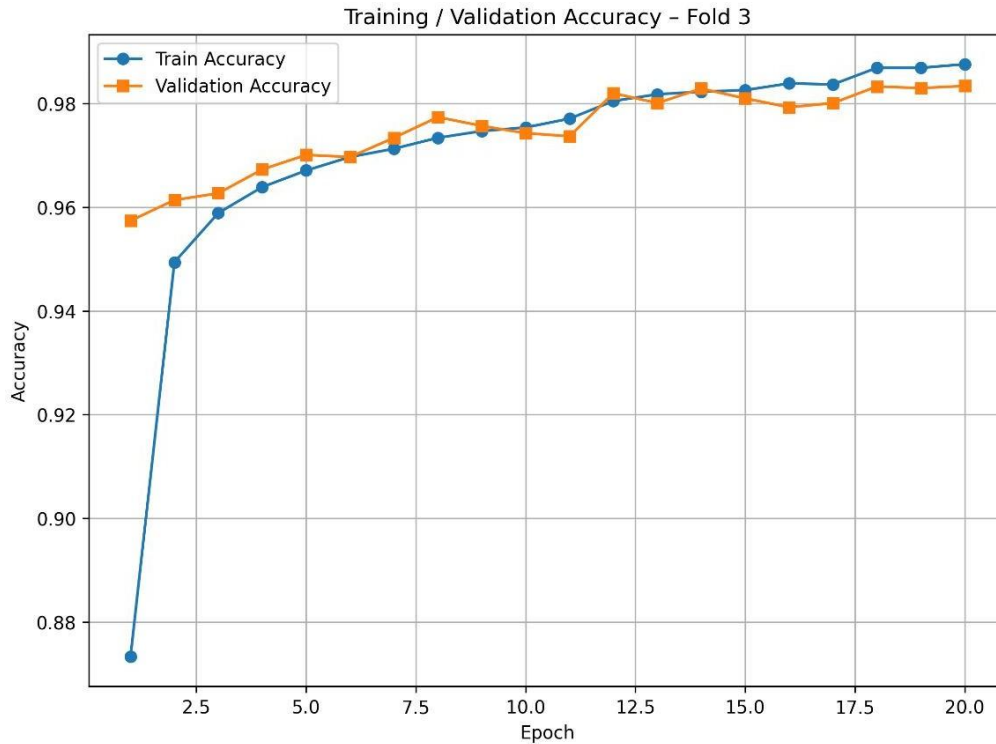


Figure 3.5: Training and validation accuracy curves of the ResNet50 model.

3.4 Software Development

This section describes the software tools and frameworks used to implement the proposed system.

- Programming language: Python
- Deep learning frameworks: TensorFlow / PyTorch
- Development environment and libraries used for data processing and model training

3.5 Future Integration Considerations (IoT and Security)

This section briefly outlines planned future enhancements, including hardware deployment and security mechanisms, which will be addressed in the second project phase.

- Planned deployment on embedded platforms
- Environmental sensor integration
- Security and data protection considerations

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